

Taking advantage of the hidden quadratic relations under a CCZ transformation

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WCC 2026
Paris

Overview

- ▶ We consider vectorial functions $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$, for $\mathbb{F}_q$ finite field
- ▶ We focus on the CCZ equivalence transformation

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- ▣Calderini et al., 2025 A new multivariate primitive from CCZ equivalence
 - ▣Caminata et al., 2026 Cryptanalysis of a multivariate CCZ scheme
 - ▣Beierle and Felke, 2025 Not Easy to Prepara a Pesto: Cryptanalysis of a Multivariate Public-Key Scheme from CCZ Equivalence

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- ▶ We use it also to characterise and study CCZ equivalent vectorial functions

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Vectorial function $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$

- ▶ we consider F in ANF, so $F = [f_1, \dots, f_m]$ with

$$f_i(z) = f_i(z_1, \dots, z_n) = \sum_{\substack{v=(v_1, \dots, v_n) \\ 0 \leq v_i < q}} a_v z_1^{v_1} \cdots z_n^{v_n}, \quad a_v \in \mathbb{F}_q$$

- ▶ f_i is a *coordinate function* of F
- ▶ *algebraic degree*

$$\deg(f_i) = \max\{\sum_{i=1}^n v_i \text{ s.t. } a_v \neq 0\}$$

$$\deg(F) = \max_i \deg(f_i)$$

Equivalence notions $F, G : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$

- affine equivalence: $\exists A_1, A_2$ affine bij. of $\mathbb{F}_q^m, \mathbb{F}_q^n$ resp. s.t.

$$G = A_1 \circ F \circ A_2$$

- EA equivalence: $\exists A : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m, \deg(A) = 1$ s.t.

$$G = A_1 \circ F \circ A_2 + A$$

- CCZ equivalence: $\exists \mathcal{A}$ affine bij. of $\mathbb{F}_q^n \times \mathbb{F}_q^m$ s.t.

$$\mathcal{A}(\Gamma_F) = \Gamma_G$$

$$\Gamma_F = \{(z, F(z)) : z \in \mathbb{F}_q^n\} \text{ graph of } F$$

- $\{\text{aff. eq.}\} \subseteq \{\text{EA eq.}\} \subseteq \{\text{CCZ eq.}\}$

CCZ equivalence

- ▶ A particular case of CCZ transformation is the t -twist:
 F and G are equivalent via a t -twist ($0 \leq t \leq \min(n, m)$) if

$$M_t(\Gamma_F) = \Gamma_G, \text{ with } M_t(x, y, u, v) = (u, y, x, v)$$

with $x, u \in \mathbb{F}_q^t$, $y \in \mathbb{F}_q^{n-t}$, $v \in \mathbb{F}_q^{m-t}$

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- ▶ CCZ transformation is completely characterized by means of EA transformations and t -twist

Indeed, the following holds:

Theorem (generalization of Canteaut&Perrin 2019 ☹ to $\mathbb{F}_q$)

If $F, G : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ are CCZ equivalent, then we can obtain G from F (and vice-versa) by composing an EA transformation, a t -twist and an EA transformation, for some $0 \leq t \leq \min(n, m)$.

Quadratic Relation

Definition

Given $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$, a *quadratic relation* for F is a quadratic function (algebraic degree at most 2) $\mathcal{R} : \mathbb{F}_q^n \times \mathbb{F}_q^m \rightarrow \mathbb{F}_q$

$$\begin{aligned}\mathcal{R}(z, w) = & \sum_{1 \leq i \leq j \leq m} \alpha_{ij} w_i w_j + \sum_{1 \leq i \leq m} \sum_{1 \leq j \leq n} \beta_{ij} w_i z_j + \sum_{1 \leq i \leq j \leq n} \gamma_{ij} z_i z_j \\ & + \sum_{1 \leq i \leq n} \delta_i z_i + \sum_{1 \leq i \leq m} \zeta_i w_i + \epsilon\end{aligned}$$

such that $\mathcal{R}(\bar{z}, F(\bar{z})) = 0$ for any $\bar{z} \in \mathbb{F}_q^n$.

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We denote with θ_F the number of linearly independent quadratic relations for F .

If $q = 2$, then $\alpha_{ii} = \gamma_{ii} = 0$.

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Theorem

If $F, G : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ are CCZ equivalent, then $\theta_F = \theta_G$.

Application in Multivariate Cryptography

classical idea: mask a simple-to-solve (secret) system $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ into a public system $\mathcal{P} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ s.t. $\mathcal{P}$ looks random (hopefully hard-to-solve)

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- ▶ we have $\theta_{\mathcal{P}} = \theta_F \geq m$ since $\{\mathcal{R}_F(z, w)\} \supseteq \{w_i - f_i(z) : 1 \leq i \leq m\}$

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- ▶ the attacker, by evaluating $\mathcal{P}$ in around $\binom{n+m+2}{2}$ inputs and solving a linear system of the same size, can recover a system of at least m quadratic relations for $\mathcal{P}$, namely $\{\mathcal{R}_i(z, w)\}$

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- ▶ the problem reduces to solve the quadratic system $\{\mathcal{R}_i(z, b) = 0\}$
- ▶ this extends the result in [1] where specific hypothesis on F and the CCZ transform were taken

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Fixed the degree of $\mathcal{P}$, the cost of this technique is polynomial in n and m :

Theorem

Given $\mathcal{P} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ CCZ equivalent to a quadratic function, with $d = \deg(\mathcal{P})$, then we can produce a system of quadratic relation for $\mathcal{P}$ with complexity $\mathcal{O}(\max\{mn^{d+2}, m^3n^d, m^6, n^6\})$.

In practical scenarios, the degree of $\mathcal{P}$ is relatively small. So for $n \geq m$, we have:

d	$\mathcal{O}(\max\{mn^{d+2}, m^3n^d, m^6, n^6\})$
3	$\mathcal{O}(n^6)$
4	$\mathcal{O}(mn^6)$
5	$\mathcal{O}(mn^7)$

Application in vectorial functions theory

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The number of linearly independent quadratic relation is a CCZ invariant.

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If $\theta_F \neq \theta_G$ then F and G are not CCZ equivalent.

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- ▶ we are also interested in determining whether a function is CCZ equivalence to a quadratic function (we say **CCZ quadratic**)

Corollary

If F is CCZ quadratic, then $\theta_F \geq m$.

On the CCZ quadratic problem

For $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$, set k_F the integer s.t. F has $q^{k_F} - 1$ affine components, $0 \leq k_F \leq m$, and let $\lambda_F = k_F(n + m + 1) - \frac{k_F(k_F+1)}{2}$.

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Proposition (necessary conditions)

If $F : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ is CCZ quadratic then:

1. $\theta_F \geq m$

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3. if $q = 2$ and $k_F = m - 1$, then $\theta_F \in \{\lambda_F + 1, \lambda_F + 3\}$

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3. if $q = 2$ and $k_F = m - 1$, then $\theta_F \in \{\lambda_F + 1, \lambda_F + 3\}$
4. if $q = 2$ and $k_F = m - 2$, then $\theta_F \in \{\lambda_F + r : r \in [2, 3, 4, 6, 7, 8]\}$

Remark

The last two results are obtained thanks to a partial classification of quadratic Boolean functions $G : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^\ell$, $\ell = 1, 2$.

Computational analysis

Performed for $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$

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- ▶ for $n = 4, \dots, 12$ identified functions not CCZ quadratic thanks to the previous results
- ▶ however, there are functions F not CCZ quadratic but s.t. $\theta_F \geq m$ (so [Condition 1.](#) in previous proposition is not sufficient)
- ▶ anyway, the pair $[\theta_F, k_F]$ can be used as a first discriminating factor for CCZ classes
 - ▶ for $n = 4$ we have a complete classification in CCZ classes

$[\theta_F, k_F]$	$\#cl.$	$[\theta_F, k_F]$	$\#cl.$	$[\theta_F, k_F]$	$\#cl.$	$[\theta_F, k_F]$	$\#cl.$
[21, 0]	3576	[21, 1]	425	[21, 2]	29	[21, 3]	1
[22, 0]	45	[22, 1]	43	[22, 2]	12	[22, 3]	2
[23, 0]	1	[23, 1]	5	[23, 2]	6	[23, 3]	1
[24, 2]	1	[24, 3]	2	[25, 3]	1	[26, 4]	1

To conclude

- ▶ We introduced the notion of *quadratic relation*, leading to
 - ▶ a technique to reduce a multivariate CCZ system back to a quadratic one
 - ▶ a new CCZ invariant, relevant for the CCZ quadratic problem
- ▶ The quadratic relations (and the value θ_F) are computable also in relatively large dimensions

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What to do next . . .

- ▶ Study further the form of the quadratic system obtained from the public system $\mathcal{P}$ with this technique
- ▶ Refine further the value θ_F for CCZ quadratic functions

Some references

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Thank you for your attention
and
thank you Giulia!

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